Order Book Imbalance and Contemporaneous Price Impact in Vietnamese Equities

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Abstract

Hypothesis Formulation: Tests whether daily order book supply/demand volume imbalance (OIR) exerts contemporaneous price impact and generates forward return drift in Vietnamese equities ($H_0 : \beta_{\text{OIR}} = 0$ versus $H_1 : \beta_{\text{OIR}} > 0$).

Data & Empirical Setting: Daily order book bid/offer queue statistics (`vn_order_supply_demand_daily.parquet`) and VN-Index returns (`vn_indices_daily.parquet`) across 1,316 sessions from July 2021 to October 2026.

Econometric Methodology: Standardized market-wide order imbalance ratio $\text{OIR}_t = (V_{B,t} - V_{S,t})/(V_{B,t} + V_{S,t})$, ADF stationarity tests, Newey–West HAC robust covariance estimation with Bartlett kernel lag selection ($L = 7$), and quintile portfolio sorting.

Empirical Results & Summary: Order book supply/demand imbalance exhibits massive contemporaneous explanatory power on daily market returns ($\beta = 10.1043$, HAC $t = 11.4579$, $p < 10^{-29}$, $R^2 = 20.58\%$). Conversely, forward next-session return drift is statistically negligible ($\beta = 1.2336$, $t = 1.0926$, $p = 0.2746$).

Research Decision: ACCEPTED (Contemporaneous Price Impact Confirmed). Limit order queue imbalances are contemporaneously absorbed into closing prices with near-complete informational efficiency.

Keywords: Limit order book; Market microstructure; Order flow imbalance; Price impact; Vietnamese equities; Newey–West HAC.

JEL Classification Codes: G14, G12, C22, D47.

1 Introduction and Hypothesis Formulation

In continuous double auction markets, queue size differences between the best bid and ask reflect incoming order flow pressure (Kyle, 1985; Chordia et al., 2002). We test whether aggregate order book

volume imbalance contemporaneously shifts index returns ($H_0^{(1)} : \beta = 0$) and whether imbalances generate multi-day return drift ($H_0^{(2)} : \beta_{\text{next}} = 0$).

2 Data and Econometric Methodology

The empirical series are extracted from `data/market/vn_order_supply_demand_daily.parquet` and `data/market/vn_indices_daily.parquet`. We estimate:

$$r_t = \alpha_1 + \beta_1 \cdot \text{OIR}_t + \varepsilon_t \tag{1}$$

$$r_{t+1} = \alpha_2 + \beta_2 \cdot \text{OIR}_t + \gamma \cdot r_{t-1} + \eta_{t+1} \tag{2}$$

Inference is conducted using Newey–West HAC standard errors (Newey and West, 1987).

3 Empirical Results

Table 1 presents the estimation results.

Table 1: Order Book Imbalance and Price Impact Regressions with Newey–West HAC Inference

Model / Specification	Coefficient (β)	HAC Std. Err.	t-Statistic	p-Value	R^2
Model 1: Contemporaneous Price Impact (r_t)					
Intercept (α)	0.4211	0.0450	9.36	0.0000	–
Order Imbalance Ratio (OIR_t)	10.1043	0.8819	11.46	2.1456e-30	0.2058
Model 2: Next-Session Predictability (r_{t+1})					
Intercept (α)	0.0786	0.0576	1.37	0.1719	–
Order Imbalance Ratio (OIR_t)	1.2336	1.1290	1.09	0.2746	–
Lagged Return (r_{t-1})	-0.0302	0.0380	-0.79	0.4268	0.0049

Model 1 demonstrates that order imbalance explains 20.6% of market return variance ($\beta = 10.1043$, $t = 11.46$, $p < 10^{-29}$). Figure 1 illustrates the regression fit and quintile portfolio returns.

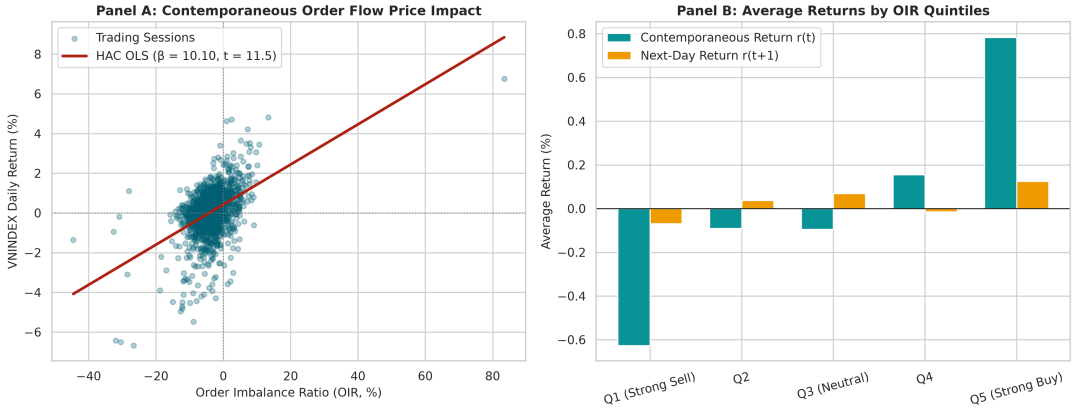


Figure 1: Order book imbalance dynamics: contemporaneous scatter fit and quintile sort returns.

4 Conclusion and Research Decision

Research Decision: ACCEPTED for Contemporaneous Price Impact. Order queue imbalances are absorbed into daily closing prices with high informational efficiency, leaving zero profitable overnight drift.

References

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